

THUMMIM MEKURIA

PhD Candidate, Astronomy & Astrophysics — University of California, Santa Cruz

tmekuria@ucsc.edu | linkedin.com/in/thummim-mekuria | thummim-mekuria.github.io

Astrophysics PhD candidate specializing in multi-tracer analysis of star-forming gas and galactic outflows, combining hydrodynamic simulations, radio/spectral-line observations, and computational pipelines to study the physics of galaxy and star formation from high-redshift systems to local molecular clouds.

EDUCATION

PhD Candidate, Astronomy & Astrophysics <i>University of California, Santa Cruz — Advisor: Dr. Nia Imara</i>	Expected 2027
M.Sc. in Astronomy & Astrophysics <i>University of California, Santa Cruz</i>	Dec 2025
B.A. in Astrophysics & Philosophy (Minor: Studio Art) <i>Pomona College — Advisors: Dr. Laura Perini and Dr. Thomas Moore — Awards: Distinction in Senior Exercise (2021), Pomona College Scholar (2017, 2021)</i>	May 2021

RESEARCH EXPERIENCE

Graduate Student Researcher <i>University of California, Santa Cruz</i>	Sep 2022 - Present
- Conduct multi-tracer analysis of spectral line emission at radio frequencies (21-cm, CO rotational lines), matching spatial and spectral resolution across datasets - Segment hyperspectral datasets into cloud catalogs to quantify the physical properties of molecular and atomic gas - Correlate the three-dimensional spatial distribution and kinematics of star-forming regions - Develop analytical models for the co-evolution of molecular and atomic gas in stellar nurseries	
Research Analyst, Center for Computational Astrophysics <i>Flatiron Institute</i>	Jan 2025 - May 2025
- Designed a numerical experiment investigating the physics and initial conditions leading to the formation of high-redshift, UV-bright galaxies discovered by JWST - Analyzed outputs from 3D radiation-hydrodynamic simulations (FLASH), measuring mass, momentum, and energy-loading factors of feedback-driven winds - Built parallel-processing pipelines to condense, analyze, and visualize simulation outputs	
LISA Simulation Research Assistant <i>Pomona College</i>	Jun 2021 - Aug 2021
- Tested and implemented adaptive time-stepping algorithms (Runge-Kutta, leapfrog) to build a module modeling the spin angular momentum evolution of a stellar binary - Incorporated the spin module into a simulation of gravitational wave detection by LISA - Updated and reorganized the existing codebase, improving efficiency and reducing runtime - Authored code documentation and a project manual synthesizing 10 years of student contributions, still used today to train incoming research assistants	
Laboratory Coordinator <i>Pomona College</i>	Sep 2021 - Jun 2022
- Calibrated an echelle spectrograph mounted on Brackett Observatory and gathered preliminary data - Prepared in-class demonstrations and worksheets for weekly laboratory experiments - Organized meetings for racial and gender minorities within the Physics and Astronomy department	
Near-Earth Object Observer <i>Jet Propulsion Laboratory</i>	Aug 2019 - Mar 2020
- Remotely operated a telescope at Table Mountain Observatory to follow up on and constrain trajectories of 50+ asteroids and near-Earth objects - Planned observing runs using ephemerides from JPL Horizons and the Minor Planet Catalog	
Big Data Intern <i>DARA / IAU Office of Astronomy for Development</i>	Jun 2020 - Aug 2020
- Collected and cleaned social network data, categorized by affective qualities, for use in a high-school machine-learning curriculum - Developed a year-long research curriculum for the OAD Flagship Project on Astronomy and Mental Health	

TEACHING & MENTORING

Research Mentor, University of California, Santa Cruz	
Mentor — Characterizing Supernova-Driven Outflows via Sonification (Jan 2026 - present): designed curriculum for the ASTR09 project teaching undergraduates research skills including programming, data analysis, and scientific presentation Co-Mentor — Atmospheric Turbulence Site Monitoring (Jan - Jun 2024): wrote Jupyter notebooks and Python scripts to teach students image reduction techniques	
Teaching Assistant, University of California, Santa Cruz	
- ASTR 01: Intro to the Cosmos (Sep - Dec 2025) - ASTR 127: Physics of Stars (Apr - Jun 2023)	
Peer Mentor and Writing Partner, Pomona College	Aug 2018 - May 2021
General Relativity, Classical Mechanics, Intro Computer Science in Python, Electricity & Magnetism, Philosophy of Physics, History of Science & Technology	

PUBLICATIONS

Mekuria, T. & Imara, N. (2026). "Kinematics of HI Envelopes Associated with Molecular Clouds." *Astronomical Journal*, 171, 200.

Mekuria, T. et al. "Outflows are Driven by Supernovae in a $z = 10$ UV-bright Galaxy." *In prep.*

CONFERENCE PRESENTATIONS

- NSBP & NSHP Joint Conference — Nov 2025
- National Society of Black Physicists Conference — Nov 2021
- Conference for Undergraduate Women in Physics — Oct 2021
- "Developing an Algorithm to Model the Spin Evolution of a Binary Star System," Research Symposium — May 2025

SKILLS

Research: Segmenting hyperspectral datasets into cloud catalogs; concurrent analysis of multi-tracer observations; fitting models to data; developing analytical models from empirical measurements

Computing: Building parallel-computing pipelines using NumPy, Pandas, and Astropy; writing Jupyter notebook tutorials and interactive data-visualization/sonification modules

Observing: Remote and in-person astronomical observing and observation planning

Communication: Written and verbal science communication for technical and general audiences

PROGRAMMING LANGUAGES	NATURAL LANGUAGES
Python (Advanced) · Xojo (Proficient) · Fortran (Intermediate) · IDL (Beginner)	Amharic (Fluent) · English (Fluent) · French (Fluent) · Arabic (Intermediate)